

Decision and Risk Analysis

Lecture 5: **Bayesian Updating and Uncertainty Propagation**

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Example

A company has deployed a new ETL pipeline.

Let

$$\pi = \Pr(\text{a randomly selected record passes all validation checks}).$$

You are performing a quality audit.

The goal is to estimate the unknown pass rate π as validation results arrive in batches.

Three analysts begin with different prior beliefs about π .

Analyst	Prior	Interpretation	Prior mean
A: Optimistic	Beta(18, 2)	Pipeline was heavily tested	0.90
B: Neutral	Beta(1, 1)	Uniform prior; little prior info	0.50
C: Skeptical	Beta(3, 7)	Similar releases have had issues	0.30

Before seeing the data, each analyst:

What proportion of records do you expect to pass validation, and how confident are you?

Bayesian Updating Rule

Suppose

$$\pi \sim \text{Beta}(\alpha, \beta),$$

and

$$Y \mid \pi \sim \text{Binomial}(n, \pi).$$

After observing y passing records out of n checked records,

$$\pi \mid y \sim \text{Beta}(\alpha + y, \beta + n - y).$$

- ▶ α represents prior weight associated with passing records.
- ▶ β represents prior weight associated with failing records.
- ▶ $\alpha + \beta$ represents the effective strength of the prior.

Batch 1: First Audit Results

Inspect 10 records:

$$y_1 = 6, \quad n_1 = 10.$$

Therefore, the observed pass rate is

$$\frac{6}{10} = 0.60.$$

Each analyst updates their prior according to

$$\text{Beta}(\alpha, \beta) \longrightarrow \text{Beta}(\alpha + 6, \beta + 4).$$

Batch 1: Posterior Results

Analyst	Prior	Posterior	Posterior mean
A	Beta(18, 2)	Beta(24, 6)	0.800
B	Beta(1, 1)	Beta(7, 5)	0.583
C	Beta(3, 7)	Beta(9, 11)	0.450

All three analysts observed exactly the same data.

Why did they not all conclude that $\pi = 0.60$?

The posterior mean can be written as

$$\mathbb{E}(\pi \mid y) = \frac{\alpha + \beta}{\alpha + \beta + n} \mathbb{E}(\pi) + \frac{n}{\alpha + \beta + n} \frac{y}{n}.$$

Why? (the Posterior Mean Is a Weighted Mean)

Start with the Beta-Binomial posterior:

$$\pi \mid y \sim \text{Beta}(\alpha + y, \beta + n - y).$$

Therefore,

$$\mathbb{E}(\pi \mid y) = \frac{\alpha + y}{\alpha + \beta + n}.$$

Rewrite the numerator as two pieces:

$$\frac{\alpha + y}{\alpha + \beta + n} = \frac{\alpha}{\alpha + \beta + n} + \frac{y}{\alpha + \beta + n}.$$

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Multiply and divide each piece by a useful quantity:

$$\frac{\alpha}{\alpha + \beta + n} = \frac{\alpha + \beta}{\alpha + \beta + n} \frac{\alpha}{\alpha + \beta}, \quad \text{and} \quad \frac{y}{\alpha + \beta + n} = \frac{n}{\alpha + \beta + n} \frac{y}{n}.$$

Putting the pieces together,

$$\mathbb{E}(\pi | y) = \frac{\alpha + \beta}{\alpha + \beta + n} \frac{\alpha}{\alpha + \beta} + \frac{n}{\alpha + \beta + n} \frac{y}{n}.$$

Since

$$\mathbb{E}(\pi) = \frac{\alpha}{\alpha + \beta},$$

we get

$$\mathbb{E}(\pi | y) = \underbrace{\frac{\alpha + \beta}{\alpha + \beta + n}}_{\text{prior weight}} \underbrace{\mathbb{E}(\pi)}_{\text{prior mean}} + \underbrace{\frac{n}{\alpha + \beta + n}}_{\text{data weight}} \underbrace{\frac{y}{n}}_{\text{sample proportion}}$$

And the two weights add to one!

$$\frac{\alpha + \beta}{\alpha + \beta + n} + \frac{n}{\alpha + \beta + n} = 1.$$

The Posterior as a Weighted Compromise

The posterior mean can be written as

$$\mathbb{E}(\pi \mid y) = \frac{\alpha + \beta}{\alpha + \beta + n} \mathbb{E}(\pi) + \frac{n}{\alpha + \beta + n} \frac{y}{n}.$$

Thus, the posterior mean is a weighted average of

- ▶ the prior mean $\mathbb{E}(\pi)$, and
- ▶ the observed sample proportion y/n .

The quantity $\alpha + \beta$ acts as an **effective prior sample size**.

Batch 1: Interpreting Prior Strength

Analyst A has a relatively strong prior:

$$\alpha + \beta = 18 + 2 = 20.$$

The new dataset contains only

$$n = 10$$

observations.

Therefore, the posterior for Analyst A remains relatively close to the prior belief.

Analyst B has a much weaker prior:

$$\alpha + \beta = 1 + 1 = 2.$$

Therefore, the observed data have much greater influence on the posterior.

Sequential Bayesian Learning

The audit continues.

Batch 2 contains 40 additional records:

30 PASS, 10 FAIL.

What should be used as the prior before analyzing Batch 2?

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The Batch 1 posterior becomes the Batch 2 prior.

Batch 2: Sequential Updating

Each analyst uses the Batch 1 posterior as the new prior.

Analyst A: $\text{Beta}(24, 6) \rightarrow \text{Beta}(54, 16)$,

Analyst B: $\text{Beta}(7, 5) \rightarrow \text{Beta}(37, 15)$,

Analyst C: $\text{Beta}(9, 11) \rightarrow \text{Beta}(39, 21)$.

Posterior means:

0.771, 0.712, 0.650.

The analysts still disagree, but their posterior beliefs are becoming more similar.

What Is Uncertainty Propagation?

We do not know the true pass rate π . A Beta distribution describes which values of π are plausible and how strongly we believe them.

When a new batch arrives, we update the **entire distribution**:

$$\underbrace{\text{Beta}(24, 6)}_{\text{after Batch 1}} \xrightarrow{30 \text{ passes, } 10 \text{ failures}} \underbrace{\text{Beta}(54, 16)}_{\text{after Batch 2}}.$$

Propagation means carrying that uncertainty forward: the posterior from one batch becomes the prior for the next batch.

What Changes as Uncertainty Propagates?

For Analyst A, the estimated pass rate and its uncertainty both change.

	Distribution for π	Mean	Standard deviation
After Batch 1	Beta(24, 6)	0.800	0.072
After Batch 2	Beta(54, 16)	0.771	0.050

The mean moves toward the new evidence. The smaller standard deviation means that plausible values of π are more tightly clustered.

Pass along the full posterior distribution, not just its mean.

Batch 3: More Data Arrive

Batch 3 contains another 150 records:

114 PASS, 36 FAIL.

Across all three batches,

$$y = 6 + 30 + 114 = 150,$$

and

$$n = 10 + 40 + 150 = 200.$$

Therefore, the overall observed pass rate is

$$\frac{150}{200} = 0.75.$$

Final Posterior Results

Analyst	Final posterior	Posterior mean
A	Beta(168, 52)	0.764
B	Beta(151, 51)	0.748
C	Beta(153, 57)	0.729

The three posterior means are now much closer together.

As the amount of observed data increases,

weight of prior $\downarrow$, weight of data $\uparrow$.

- ▶ Analyst A begins optimistic and updates downward.
- ▶ Analyst C begins skeptical and updates upward.
- ▶ Analyst B begins with a diffuse prior and responds quickly to the observed data.
- ▶ As more observations arrive, the posterior distributions become increasingly concentrated.
- ▶ The analysts' posterior beliefs move closer to the empirical pass rate of 0.75.

Sequential Updating vs. All at Once

Instead of processing the observations in three batches, suppose we analyze all 200 records simultaneously.

For Analyst A,

$$\text{Beta}(18, 2) \longrightarrow \text{Beta}(18 + 150, 2 + 50).$$

Therefore,

$$\text{Beta}(18 + 150, 2 + 50) = \text{Beta}(168, 52).$$

This is exactly the same final posterior obtained using sequential updating.

Which approach is correct?

1. Process 10 records, then 40 records, then 150 records.
2. Process all 200 records at once.

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Both approaches give the same final posterior.

Sequential updating = updating with all the data at once

Data-Order Invariance

The final posterior depends on the total number of passes and failures, not on the order in which the batches are processed.

For Analyst A,

$$\text{Prior} = \text{Beta}(18, 2).$$

The complete dataset contains

150 passes, 50 failures.

Thus,

$$\text{Beta}(18 + 150, 2 + 50) = \text{Beta}(168, 52).$$

Reversing the order of the batches changes the intermediate posteriors, but not the final posterior.

Questions

1. Why do the three analysts disagree after Batch 1 even though they observed identical data?
2. Which prior exerts the greatest influence after Batch 1, and why?
3. Why do the posterior distributions become more similar as more records are inspected?
4. Why does sequential updating produce the same final posterior as analyzing all 200 observations simultaneously?

Bayesian Learning

Prior knowledge + new evidence $\longrightarrow$ updated knowledge

Then,

Updated knowledge becomes the next prior

For the Beta–Binomial model, the mechanics are simple:

$$\text{Beta}(\alpha, \beta) \longrightarrow \text{Beta}(\alpha + y, \beta + n - y).$$

R: Batch 1

```
library(bayesrules)

plot_beta_binomial(
  alpha = 18,
  beta  = 2,
  y     = 6,
  n     = 10
)
```

This displays Analyst A's prior, likelihood information, and posterior after Batch 1.

R: Sequential Updating

After Batch 1, Analyst A has

$$\pi \mid y_1 \sim \text{Beta}(24, 6).$$

Use this posterior as the prior for Batch 2:

```
plot_beta_binomial(  
  alpha = 24,  
  beta  = 6,  
  y      = 30,  
  n      = 40  
)
```

The result is

$$\pi \mid y_1, y_2 \sim \text{Beta}(54, 16).$$

R: Batch 3

Use the Batch 2 posterior as the next prior:

```
plot_beta_binomial(  
  alpha = 54,  
  beta  = 16,  
  y      = 114,  
  n      = 150  
)
```

This produces

$$\pi \mid y_1, y_2, y_3 \sim \text{Beta}(168, 52).$$

The same posterior is obtained by analyzing all 200 observations at once.